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Physical Design

Physical Design of RISC-V based System-on-Chip using Open Lane

A Report

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1.Abstract:

This project presents the complete physical design of a RISC-V based PicoSoC using the open-source OpenLane EDA flow targeting the SkyWater 130 nm (SKY130) Process Design Kit. PicoSoC integrates the PicoRV32 processor core (RV32IMC ISA), a UART peripheral, an SPI flash controller, and an internal SRAM block, all interconnected via a 32-bit system bus. The design flow encompasses RTL simulation using Icarus Verilog, logic synthesis using Yosys, floorplanning, standard cell placement, clock tree synthesis, routing using TritonRoute, and physical sign-off verification including Magic DRC and Netgen LVS.

The OpenLane run completed successfully in 1 hour 21 minutes. The synthesized design contains 42,194 standard cells placed on a die area of 2.41 mm². Static Timing Analysis confirms timing closure at 100 MHz with a worst negative slack of 0.0 ns. Magic DRC reported zero violations and LVS returned a clean result. The total routed wire length is approximately 2.99 million microns across six metal layers with 368,965 vias. Power consumption at the typical corner is approximately 178 mW. This project demonstrates that open-source VLSI tools are capable of producing tapeout-quality physical layouts for a complete System-on-Chip design within an academic setting.

2.Objectives:

- To implement the complete physical design of a RISC-V based PicoSoC using the open-source OpenLane EDA flow targeting the *SKY130 130 nm PDK*.
- To perform RTL simulation of the PicoSoC design and verify functional correctness prior to physical implementation.
- To carry out logic synthesis using *Yosys*, mapping the RTL to standard cells from the *sky130_fd_sc_hd* library, and to optimize for area under the AREA 0 synthesis strategy.
- To complete floorplanning by defining die and core area boundaries, placing pre-hardened macro blocks (PicoRV32, UART, SPI, Memory, Memory Decoder) at optimal coordinates to minimize routing congestion.
- To construct a robust Power Distribution Network (PDN) and to perform clock tree synthesis using *TritonCTS* to achieve minimal clock skew across all sequential elements.
- To execute global and detailed routing using *FastRoute* and *TritonRoute*, targeting zero DRC violations and ensuring all signal nets are correctly connected.
- To perform physical signoff verification including Magic DRC, LVS using Netgen, antenna rule checking, and Static Timing Analysis, and to analyze the results.
- To extract and document key design metrics including cell count, wire length, via count, timing slack, power consumption, and routing layer utilization.
- To visualize the final GDS layout using *KLayout* and *OpenROAD*, and to validate that the gate-level simulation matches the RTL simulation outputs.

3.Block Diagrams:

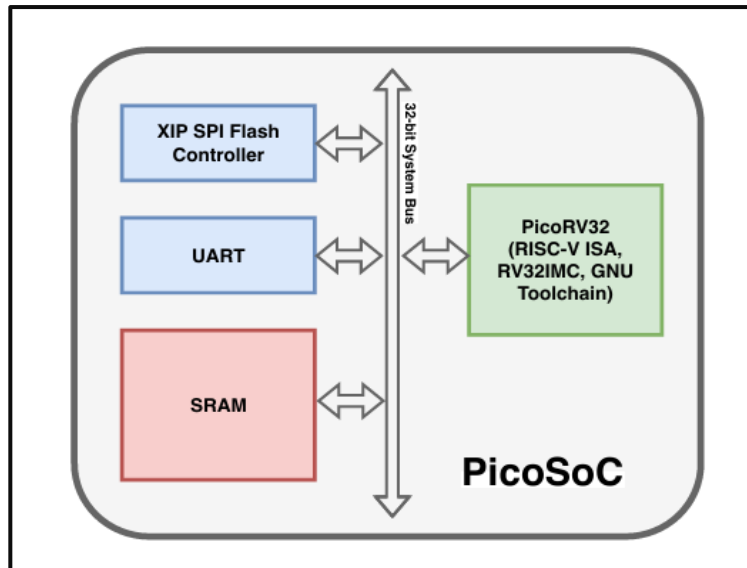


Fig.1 PicoSoC architecture (Overall)

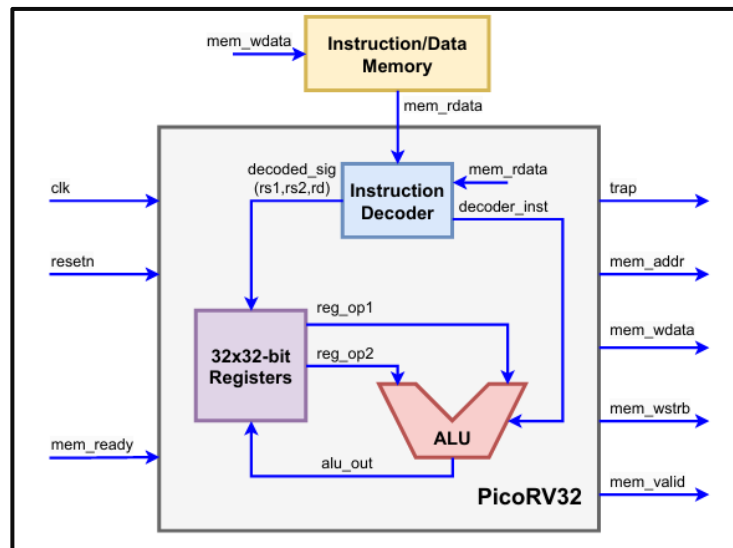


Fig. 2 PicoRV32 simplified block architecture

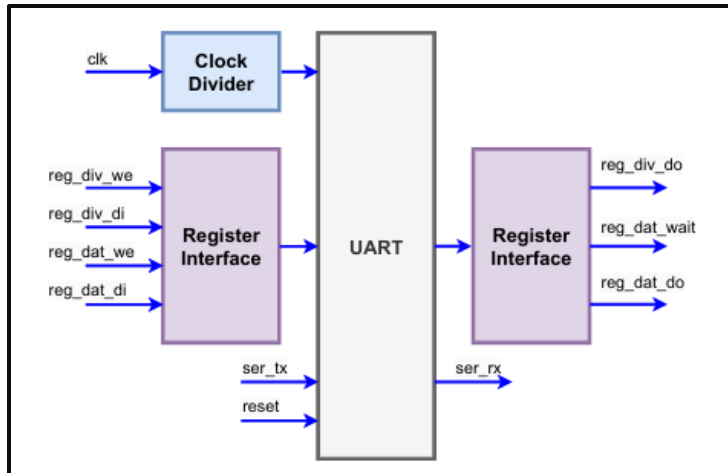


Fig. 3 UART block diagram

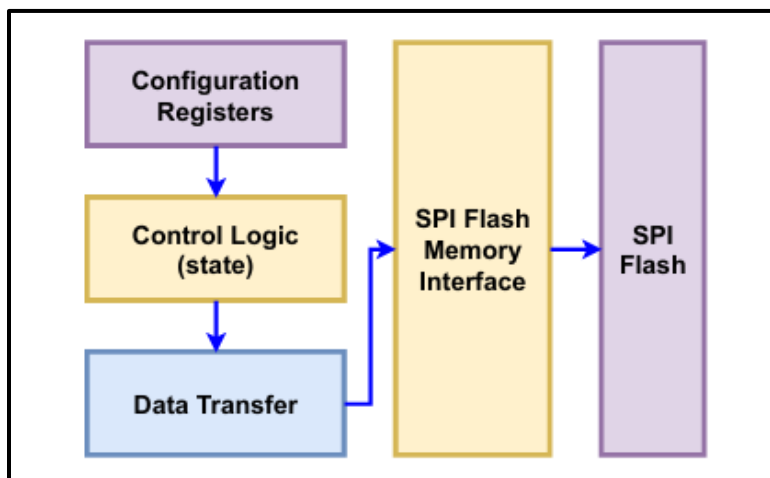


Fig. 4 SPI block diagram

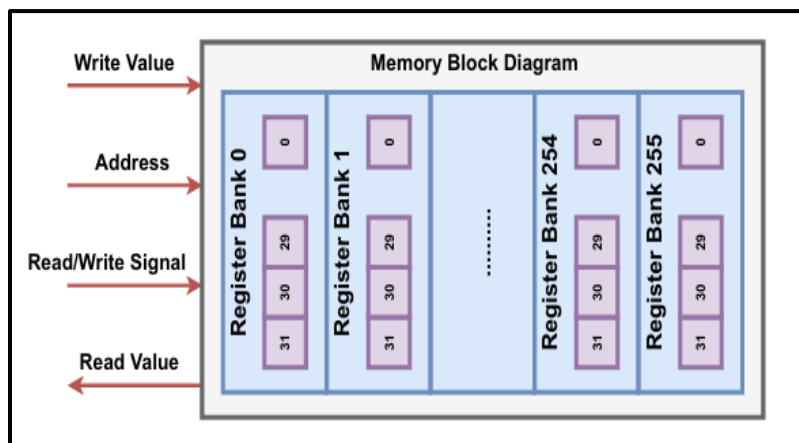


Fig. 5 Memory Block Diagram

RTL to GDS II:

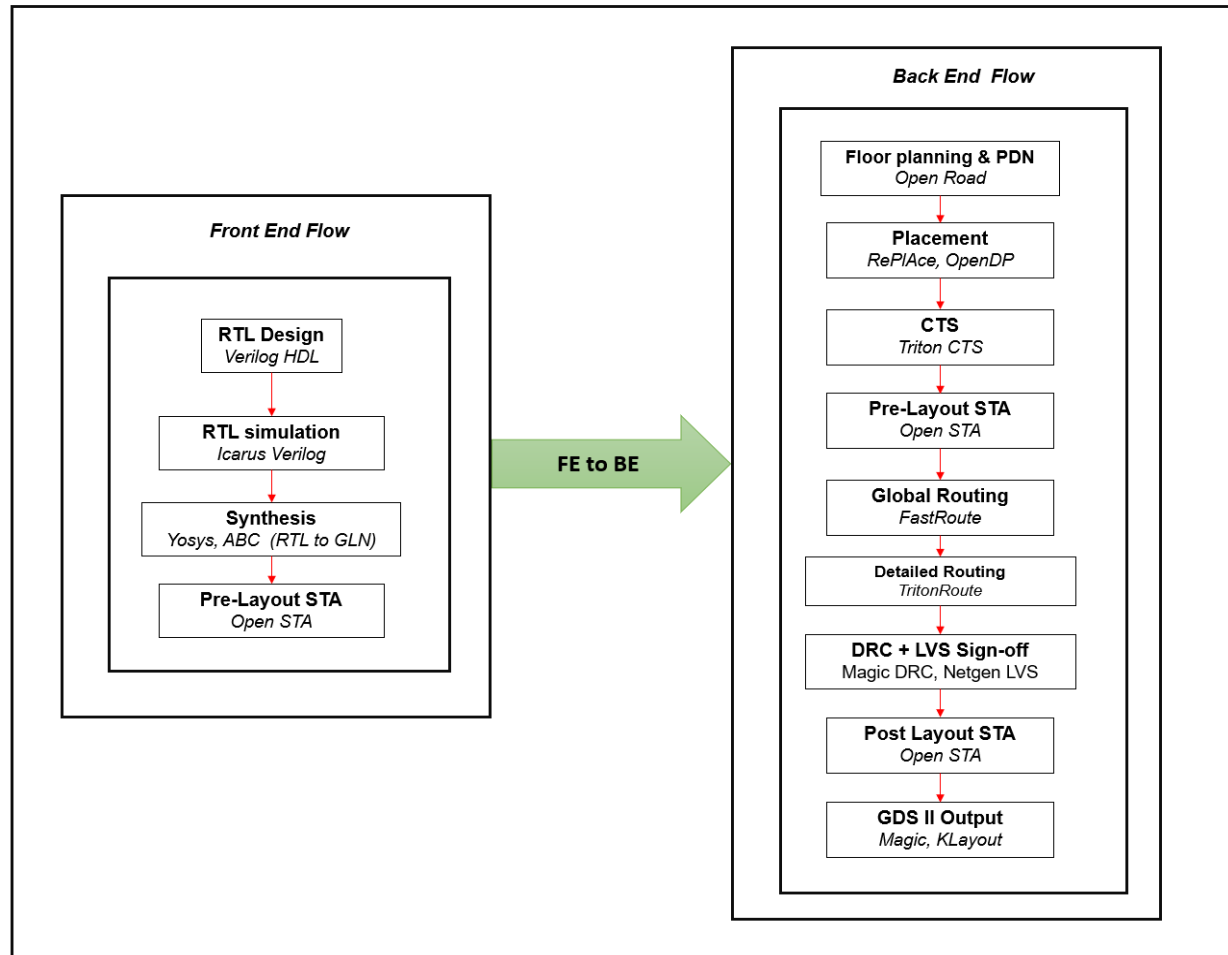


Fig .6 RTL to GDSII Physical Design Flow using Open-Lane (SKY130 PDK)

4.The Flow (Methodology):

4.1 Front-End Flow:

4.1.1 RTL-Design:

- Hardware description of PicoSoC written in Verilog HDL
- Modules designed: PicoRV32 CPU, UART, SPI Flash Controller, SRAM, and Memory Decoder
- Each module written as a separate synthesizable Verilog source file.

4.1.2 Synthesis:

- Logic synthesis performed using Yosys with ABC as the optimization engine
- RTL description mapped to a Gate-Level Netlist (GLN) using the *sky130_fd_sc_hd* standard cell library.
- Synthesis strategy set to area-optimized with hierarchy preservation enabled
- Clock period constraint applied to guide timing-driven synthesis.

4.1.3 Pre-layout STA:

- Performed using Open-STA on the synthesized gate-level netlist
- Setup and hold timing constraints verified at the target clock frequency
- Timing closure confirmed before proceeding to physical implementation

4.2 Back-End Flow:

4.2.1 Floor planning & Power Distribution Network (PDN)

- Performed using Open ROAD
- Die area and core boundaries defined based on target utilization
- Pre-hardened macro blocks placed at fixed coordinates to minimize routing congestion
- Power Distribution Network constructed with horizontal and vertical power straps on upper metal layers
- Core utilization set conservatively to ensure sufficient routing headroom

4.2.2 Placement

- Global placement performed using RePlAce to minimize wirelength
- Detailed placement performed using OpenDP to legalize cell positions
- Antenna diode insertion carried out during placement to prevent gate oxide damage

4.2.3 Clock Tree Synthesis (CTS)

- Performed using TritonCTS
- Balanced clock distribution tree constructed across all sequential elements
- Clock skew minimized to meet setup and hold timing requirements

4.2.4 Global Routing

- Performed using FastRoute
- Routing guides generated for all signal nets
- Congestion adjustment applied to ensure routability in dense regions

4.2.5 Detailed Routing

- Performed using TritonRoute
- All signal nets routed across the available metal layers of the SKY130 process
- Design Rule Check (DRC) compliance enforced during routing

4.2.6 DRC + LVS Sign-off

- Design Rule Check performed using Magic to verify layout compliance with SKY130 manufacturing rules
- Layout vs. Schematic check performed using Netgen to confirm electrical equivalence between layout and netlist
- Antenna violations addressed through antenna diode insertion and metal layer jumping

4.2.7 Post-Layout Static Timing Analysis (STA)

- Final timing sign-off performed using OpenSTA on the fully routed netlist
- Setup and hold slack verified across all timing paths
- Maximum operating frequency confirmed from critical path analysis

4.2.8 GDS II Output

- Final layout exported in GDS II format using Magic
- Layout visualized and verified using KLayout

5. Results:

5.1 Floor planning & Power Distribution Network (PDN)

The floorplan was generated using Open ROAD. Standard cell rows were initialized across the core area and the power distribution network was constructed with horizontal and vertical straps on upper metal layers. The figure below shows the initialized standard cell rows after PDN creation.

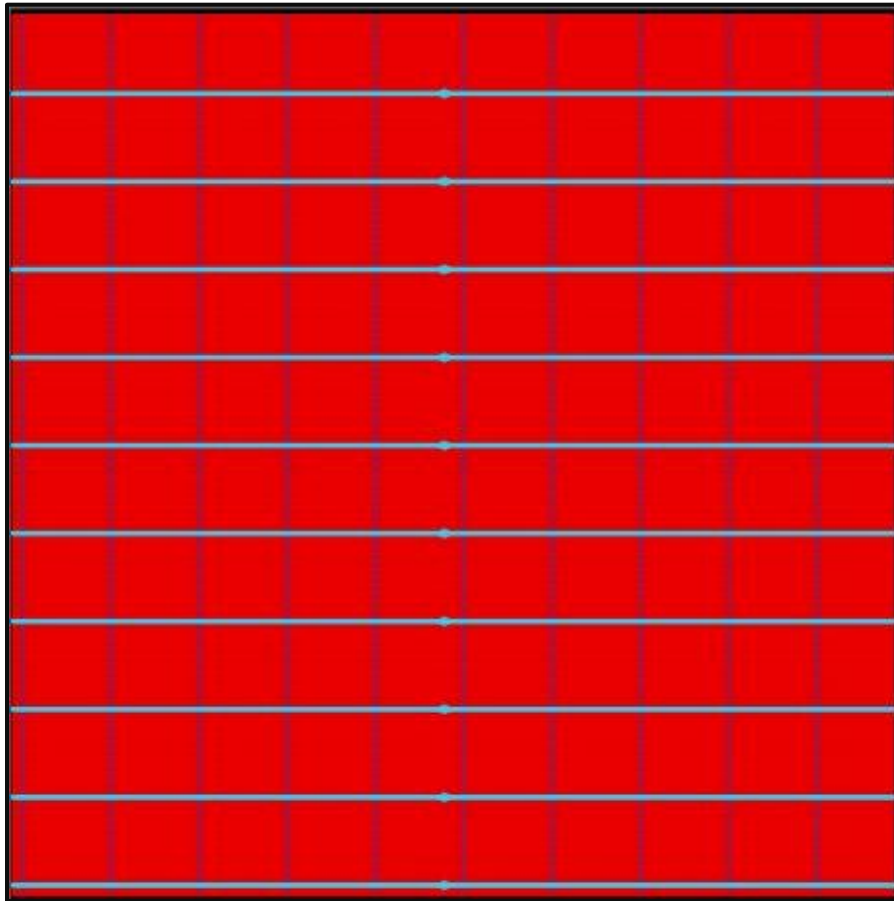


Fig .6 *Floor Planning and PDN Creation – Open ROAD*

5.2 Placement

Global placement was performed using RePlAce and detailed placement using OpenDP. The figure below shows the placement result with upper metal layers hidden in KLayout, revealing the distribution of standard cells across the active core area.

5.3 Clock Tree Synthesis:

Clock tree synthesis was performed using Triton CTS to construct a balanced clock distribution network across all sequential elements. The table below summarizes the timing metrics obtained after CTS completion.

Timing Parameter	Value
Worst Negative Slack (WNS)	0.00 ns
Total Negative Slack (TNS)	0.00 ns
Worst Setup Slack	3.76 ns
Worst Hold Slack	0.13 ns
Clock Skew	0.29 ns

Table .1 *Timing results after Clock Tree Synthesis*

5.4 Global Routing:

Global routing was performed using FastRoute to generate routing guides for all signal nets.

Metal Layer	Layer Name	Utilization (%)
Layer 1 (Local Interconnects)	li1	0.00
Layer 2	met1	33.24
Layer 3	met2	34.32
Layer 4	met3	13.01
Layer 5	met4	16.28
Layer 6	met5	1.92

Table .2 *Metal Layer Utilization after Global Routing*

5.5 Detailed Routing:

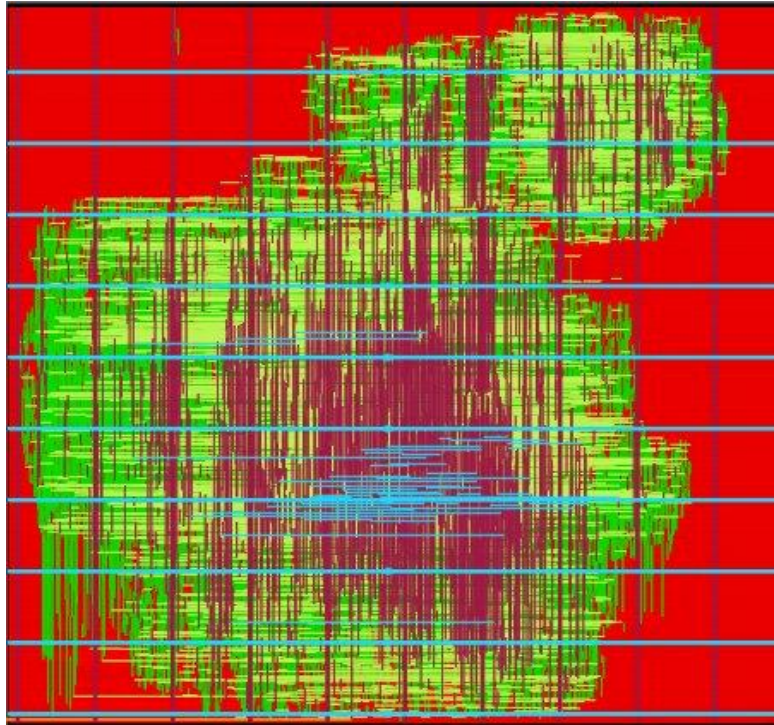


Fig .8 Complete Detailed routing – Open Road Visualization

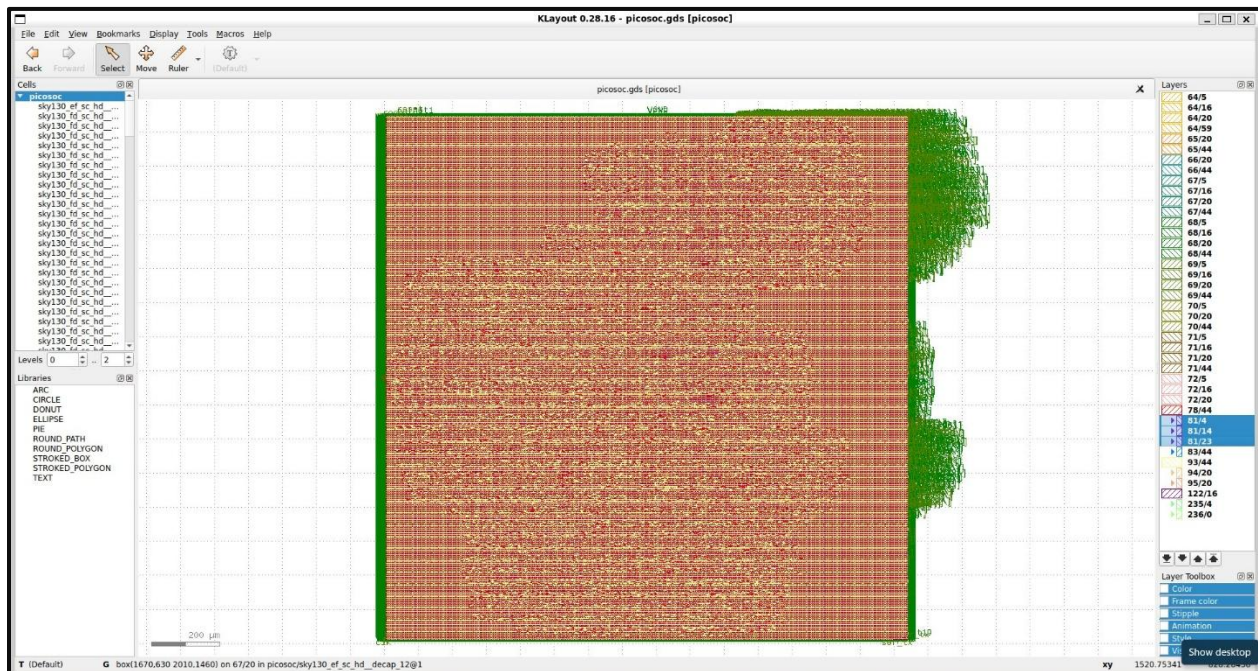


Fig .9 Routing (Upper layer hidden & lower metal layer visible)

5.6 DRC + LVS Physical Sign-off:

Physical sign-off verification was performed using Magic for Design Rule Checking (DRC) and Netgen for Layout vs. Schematic (LVS) verification. The table below summarizes the sign-off results.

Verification Check	Tool Used	Result
Design Rule Check (DRC)	Magic	0 Violations
Layout Vs Schematics	NetGen	LVS Clean
Total Nets Matched	NetGen	53,216 Nets
Antenna Pin Violations	Open ROAD	432
Antenna Net Violations	Open ROAD	326

Table .3 *Physical sign-off Verification Results – DRC & LVS*

5.7 Post Layout STA:

Final timing sign-off was performed using OpenSTA on the fully routed netlist. The table below presents the complete timing and power results at the typical process corner.

Timing Parameter	Value
Clock Period Constraint	10.00 ns
Worst Negative Slack (WNS)	0.00 ns
Total Negative Slack (TNS)	0.00 ns
Critical Path Delay	8.70 ns
Maximum Operating Frequency	100 MHz
Worst Setup Slack (Post – Route)	4.57 ns
Worst Hold Slack (Post – Route)	0.16 ns
Total Power (Typical Corner)	155mW

Table. 4 *Post – Layout STA and Power results*

5.8 GDS II Output:

- The final GDS II layout was streamed out using Magic after successful completion of all physical verification checks including DRC, LVS, and antenna sign-off.
- The layout encompasses all five macro blocks — PicoRV32 CPU, UART, SPI Flash Controller, Memory Decoder, and SRAM — along with synthesized standard cell logic and the power distribution network.
- All components are fully placed and routed within the defined die boundary of $1546.9 \times 1557.6 \mu\text{m}$ across six metal layers (li1, met1–met5).
- The complete layout was loaded and inspected in KLayout 0.28.16, confirming uniform standard cell density across the core area and correct macro placement at designated coordinates.
- The zoomed-in view reveals fine-pitch metal interconnects, via placements between adjacent metal layers, and well-defined standard cell row boundaries.

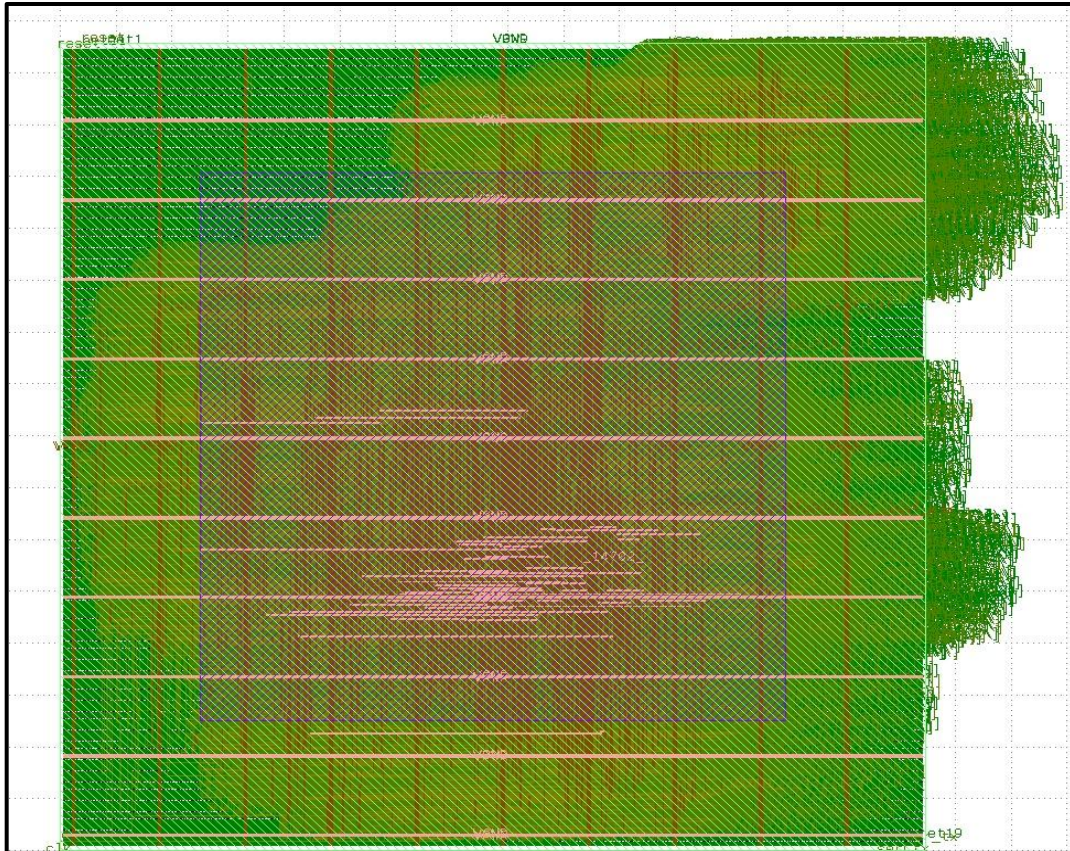


Fig .10 Complete PicoSoc GDS II Layout – Klayout 0.28.16

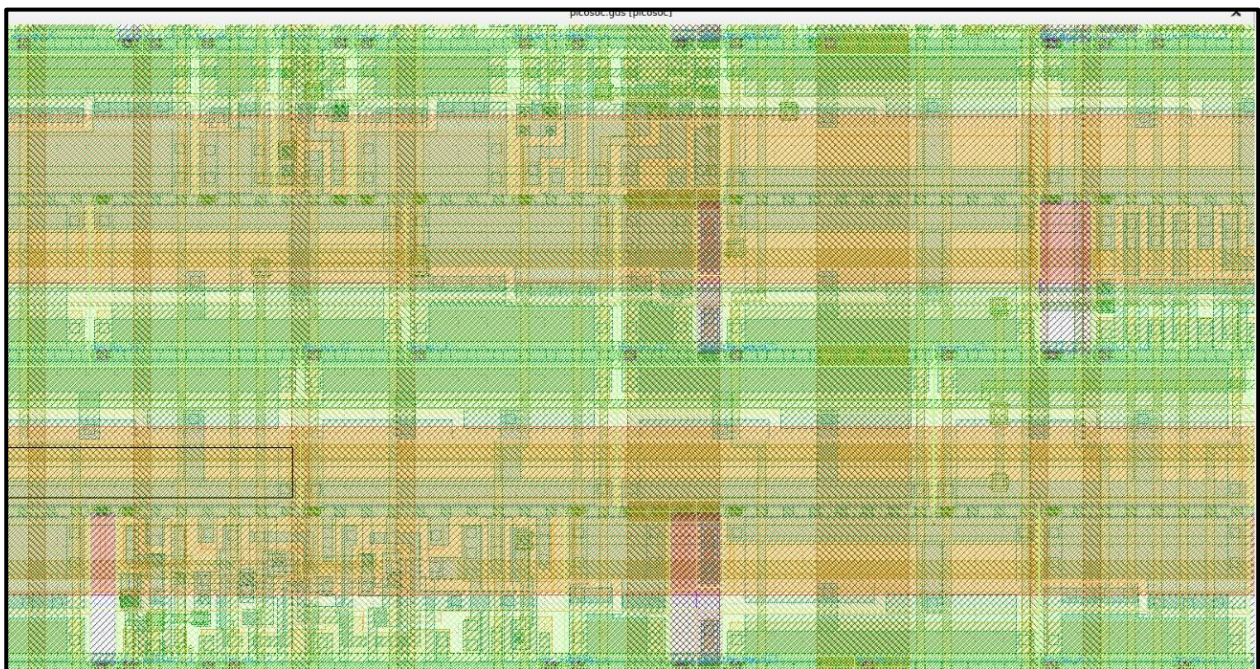


Fig .11 Zoomed-in view of PicoSoC layout showing standard cells, Metal interconnects & Vias

6. Inference:

- The OpenLane RTL-to-GDS flow completed successfully with a total runtime of 1 hour 21 minutes, confirming the viability of open-source EDA tools for complete SoC physical design.
- Full timing closure was achieved at 100 MHz with WNS = 0.0 ns and TNS = 0.0 ns, significantly exceeding the base paper's reported maximum frequency of 19 MHz [1].
- Magic DRC reported zero violations and Netgen LVS returned a clean result with 53,216 matched nets, confirming the layout is physically and electrically correct.
- TritonCTS achieved a clock skew of 0.29 ns, ensuring balanced clock distribution across all sequential elements within acceptable limits.
- Metal layers met1 and met2 carry the highest routing utilization at 33.24% and 34.32% respectively, consistent with standard cell signal routing behavior in the SKY130 process.
- Antenna violations of 432 pin and 326 net violations were detected and resolved through diode insertion, which is expected for a design of this scale.
- Total chip power at the typical corner is 155 mW, with the majority dissipated in combinational logic as expected for a processor-centric design.
- The achieved placement density of 20.43% closely matches the 20% target, confirming correct placer operation with sufficient routing headroom maintained throughout the core area.

Reference:

- [1] M. Fauzan *et al.*, "Physical Design of RISC-V based System-on-Chip using OpenLane," *2024 IEEE Asia Pacific Conference on Circuits and Systems (APCCAS)*, pp. 529–533, Nov. 2024, doi: 10.1109/APCCAS62602.2024.10808656.